

[MS-GPCAP]:

Group Policy: Central Access Policies Protocol Extension

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1 Introduction

The Group Policy: Central Access Policies Extension allows the configuring of central access policies (CAPs) on **Group Policy client** computers.

This protocol extension also contains the mechanisms that enable Group Policy administrators to retrieve policy files and configure **central access policy (CAP)** information that is stored in the **Group Policy data store**.

Sections 1.5, 1.8, 1.9, 2, and 3 of this specification are normative. All other sections and examples in this specification are informative.

1.1 Glossary

This document uses the following terms:

Active Directory: The Windows implementation of a general-purpose directory service, which uses **LDAP** as its primary access protocol. Active Directory stores information about a variety of objects in the network such as user accounts, computer accounts, groups, and all related credential information used by Kerberos [\[MS-KILE\]](#). Active Directory is either deployed as **Active Directory Domain Services (AD DS)** or Active Directory Lightweight Directory Services (AD LDS), which are both described in [\[MS-ADOD\]](#): Active Directory Protocols Overview.

Active Directory Domain Services (AD DS): A directory service (DS) implemented by a **domain controller (DC)**. The DS provides a data store for objects that is distributed across multiple **DCs**. The DCs interoperate as peers to ensure that a local change to an object replicates correctly across DCs. AD DS is a deployment of **Active Directory** [\[MS-ADTS\]](#).

Administrative tool: An implementation-specific tool, such as the Group Policy Management Console, that allows administrators to read and write **policy settings** from and to a Group Policy Object (GPO) and policy files. The Group Policy Administrative tool uses the Extension list of a GPO to determine which Administrative tool extensions are required to read settings from and write settings to the logical and physical components of a GPO.

Administrative tool extension: A Group Policy extension protocol that is identified by an Administrative tool extension GUID and invoked by a management entity such as the Group Policy Management Console. The Administrative tool extension enables the Group Policy administrator to administer policy settings associated with the specific context provided by the extension.

Administrative tool extension GUID: A GUID that enables a specific **Administrative tool extension** to be associated with settings that are stored in a GPO on the Group Policy server for that particular extension. The GUID enables the **Administrative tool** to identify the extension protocol for which settings are to be administered.

attribute: A characteristic of some object or entity, typically encoded as a name/value pair.

Augmented Backus-Naur Form (ABNF): A modified version of Backus-Naur Form (BNF), commonly used by Internet specifications. ABNF notation balances compactness and simplicity with reasonable representational power. ABNF differs from standard BNF in its definitions and uses of naming rules, repetition, alternatives, order-independence, and value ranges. For more information, see [\[RFC5234\]](#).

central access policy (CAP): An authorization policy that is specified by a GPO component and applied to policy targets to facilitate centralized access control of resources.

central access policy (CAP) object: An object stored in an LDAP directory service, such as Active Directory, that contains one or more central access rules (CARs), which specify the details of an authorization policy.

central access rule (CAR): An object that is stored in the Central Access Policy Rules List of a central access policy (CAP) object. Each CAR contains an authorization policy that specifies the resources, users, and access conditions to which the rule applies.

client-side extension (CSE): A Group Policy extension that resides locally on the **Group Policy client** and is identified by a **client-side extension GUID (CSE GUID)**.

client-side extension GUID (CSE GUID): A **GUID** that enables a specific client-side extension on the **Group Policy client** to be associated with policy data that is stored in the logical and physical components of a Group Policy Object (GPO) on the Group Policy server, for that particular extension.

computer-scoped Group Policy Object path: A scoped **Group Policy Object (GPO) path** that ends in "\\Machine".

core Group Policy engine: The software entity that implements the Group Policy: Core Protocol [\[MS-GPOL\]](#). The core Group Policy engine issues the message sequences that result in core protocol network traffic during policy application on **Group Policy clients**. The engine handles functions on behalf of the core protocol such as the Group Policy refresh interval, GPO and policy file access, GPO filtering and ordering, and invoking transport protocols for retrieving and storing policy settings.

domain: A set of users and computers sharing a common namespace and management infrastructure. At least one computer member of the set has to act as a **domain controller (DC)** and host a member list that identifies all members of the domain, as well as optionally hosting the **Active Directory** service. The domain controller provides authentication of members, creating a unit of trust for its members. Each domain has an identifier that is shared among its members. For more information, see [\[MS-AUTHSOD\]](#) section 1.1.1.5 and [\[MS-ADTS\]](#).

domain controller (DC): The service, running on a server, that implements **Active Directory**, or the server hosting this service. The service hosts the data store for objects and interoperates with other **DCs** to ensure that a local change to an object replicates correctly across all **DCs**. When **Active Directory** is operating as **Active Directory Domain Services (AD DS)**, the **DC** contains full NC replicas of the configuration naming context (config NC), schema naming context (schema NC), and one of the domain NCs in its forest. If the **AD DS DC** is a global catalog server (GC server), it contains partial NC replicas of the remaining domain NCs in its forest. For more information, see [\[MS-AUTHSOD\]](#) section 1.1.1.5.2 and [\[MS-ADTS\]](#). When **Active Directory** is operating as Active Directory Lightweight Directory Services (AD LDS), several AD LDS **DCs** can run on one server. When **Active Directory** is operating as **AD DS**, only one **AD DS DC** can run on one server. However, several AD LDS **DCs** can coexist with one **AD DS DC** on one server. The AD LDS **DC** contains full NC replicas of the config NC and the schema NC in its forest. The domain controller is the server side of Authentication Protocol Domain Support [\[MS-APDS\]](#).

domain name: The name given by an administrator to a collection of networked computers that share a common directory. Part of the domain naming service naming structure, domain names consist of a sequence of name labels separated by periods.

Domain Name System (DNS): A hierarchical, distributed database that contains mappings of **domain names** to various types of data, such as IP addresses. DNS enables the location of computers and services by user-friendly names, and it also enables the discovery of other information stored in the database.

globally unique identifier (GUID): A term used interchangeably with universally unique identifier (UUID) in Microsoft protocol technical documents (TDs). Interchanging the usage of these terms does not imply or require a specific algorithm or mechanism to generate the value.

Specifically, the use of this term does not imply or require that the algorithms described in [\[RFC4122\]](#) or [\[C706\]](#) have to be used for generating the GUID. See also universally unique identifier (UUID).

Group Policy: A mechanism that allows the implementer to specify managed configurations for users and computers in an **Active Directory** service environment.

Group Policy administrator: A **domain** administrator who is responsible for defining **policy settings** and managing the **Group Policy** infrastructure of a **domain**.

Group Policy client: A client computer that receives and applies settings of a GPO. The Group Policy client can use client-side extensions to extend the functionality of the Group Policy protocols.

Group Policy data store: A data store that consists of two types of stores. One is a physical (file system) data store on the **Group Policy file share** that contains policy settings (extension and administrative template data), which can be locally or remotely accessed depending on location. The other is a logical data store that is part of **Active Directory** and serves as a repository for GPOs that are accessible via **Lightweight Directory Access Protocol (LDAP)**.

Group Policy extension: A protocol that extends the functionality of Group Policy. Group Policy extensions consist of client-side extensions and **Administrative tool extensions**. They provide settings and other Group Policy information that can be read from and written to **Group Policy data store** components. Group Policy Extensions depend on the Group Policy: Core Protocol, via the **core Group Policy engine**, to identify GPOs containing a list of extensions that apply to a particular **Group Policy client**.

Group Policy extension GUID: A GUID that identifies a Group Policy Extension, such as a CSE or Administrative tool extension. Group Policy extension GUIDs are contained in an extension list that is an attribute of a GPO that applies to a particular Group Policy client.

Group Policy file share: A file system storage location that contains policy settings that include extension settings and Group Policy template settings for GPOs. The latter settings consist of security and registry settings, script files, and application installation information.

Group Policy Object (GPO) path: A **domain**-based Distributed File System (DFS) path for a directory on the server that is accessible through the DFS/SMB protocols. This path will always be a Universal Naming Convention (UNC) path of the form: "\\<dns domain name>\sysvol\<dns domain name>\policies\<gpo guid>", where <dns domain name> is the **DNS domain name** of the **domain** and <gpo guid> is a Group Policy Object (GPO) GUID.

Group Policy server: A server holding a database of Group Policy Objects (GPOs) that can be retrieved by other machines. The Group Policy server has to be a **domain controller (DC)**.

Lightweight Directory Access Protocol (LDAP): The primary access protocol for **Active Directory**. Lightweight Directory Access Protocol (LDAP) is an industry-standard protocol, established by the Internet Engineering Task Force (IETF), which allows users to query and update information in a directory service (DS), as described in [MS-ADTS]. The Lightweight Directory Access Protocol can be either version 2 [\[RFC1777\]](#) or version 3 [\[RFC3377\]](#).

policy application: The protocol exchange by which a client obtains all of the Group Policy Object (GPO) and thus all applicable Group Policy settings for a particular policy target from the server, as specified in [MS-GPOL]. Policy application can operate in two modes, user policy and computer policy.

policy setting: A statement of the possible behaviors of an element of a domain member computer's behavior that can be configured by an administrator.

policy target: A user or computer account for which policy settings can be obtained from a server in the same domain, as specified in [MS-GPOL]. For user policy mode, the policy target is a user account. For computer policy mode, the policy target is a computer account.

schema: The set of **attributes** and object classes that govern the creation and update of objects.

scope of management (SOM): An **Active Directory** site, **domain**, or organizational unit container. These containers contain user and computer accounts that can be managed through **Group Policy**. These **SOMs** are themselves associated with Group Policy Objects (GPOs), and the accounts within them are considered by the Group Policy Protocol [MS-GPOL] to inherit that association.

security identifier (SID): An identifier for security principals that is used to identify an account or a group. Conceptually, the **SID** is composed of an account authority portion (typically a **domain**) and a smaller integer representing an identity relative to the account authority, termed the relative identifier (RID). The **SID** format is specified in [MS-DTYP] section 2.4.2; a string representation of **SIDs** is specified in [MS-DTYP] section 2.4.2 and [MS-AZOD] section 1.1.1.2.

system volume (SYSVOL): A shared directory that stores the server copy of the **domain's** public files that has to be shared for common access and replication throughout a domain.

UTF-8: A byte-oriented standard for encoding Unicode characters, defined in the Unicode standard. Unless specified otherwise, this term refers to the UTF-8 encoding form specified in [UNICODE5.0.0/2007] section 3.9.

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [RFC2119]. All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[MS-ADA2] Microsoft Corporation, "[Active Directory Schema Attributes M](#)".

[MS-ADSC] Microsoft Corporation, "[Active Directory Schema Classes](#)".

[MS-ADTS] Microsoft Corporation, "[Active Directory Technical Specification](#)".

[MS-DTYP] Microsoft Corporation, "[Windows Data Types](#)".

[MS-GPOL] Microsoft Corporation, "[Group Policy: Core Protocol](#)".

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>

[RFC2251] Wahl, M., Howes, T., and Kille, S., "Lightweight Directory Access Protocol (v3)", RFC 2251, December 1997, <https://www.rfc-editor.org/info/rfc2251>

1.2.2 Informative References

[MS-FASOD] Microsoft Corporation, "[File Access Services Protocols Overview](#)".

[MS-GPOD] Microsoft Corporation, "[Group Policy Protocols Overview](#)".

1.3 Overview

The Group Policy: Central Access Policies Extension is a **Group Policy extension** that enhances the functionality of **Group Policy**. It enables Group Policy administrators to specify CAPs on Group Policy servers that are to be configured on a **Group Policy client** computer, such as a file server, for control of access to resources on those computers. CAPs are only live after they are applied to resources on Group Policy client computers by a local resource administrator.

Policy settings for the Group Policy: Central Access Policies Extension are specified by one or more Group Policy Objects (GPOs) that reside in the **Group Policy data store**. Each GPO contains a logical component in **Active Directory** and a physical (file system) component that is stored on a file share, such as the **Group Policy file share**, which is either remote or local to the **Group Policy server**. The logical component defines policy metadata that is held by GPO **attributes** and is used to define such things as the extensions that apply to a client and the file system location where policy settings and other information is stored. The physical component holds a specially formatted file containing identifiers that enable an implementation to locate **CAP objects** in Active Directory, to facilitate the subsequent configuration of authorization policies on Group Policy client computers. The **Group Policy administrator** uses these components to define the **central access policy (CAP)** configuration that is applied to a **policy target**, such as a Group Policy client.

The Group Policy: Central Access Policies Extension protocol implements both a client-side and an administrative-side extension, the globally unique identifiers (GUIDs) for which are specified in section 1.9. The administrative side, sometimes referred to as an administrative plug-in, is invoked by the **Administrative tool** when the Group Policy administrator creates, modifies, or deletes central access policies. The client side, sometimes referred to as a client plug-in, is invoked to initiate the application of client access policies on a target computer, such as a Group Policy client.

1.3.1 Background

The Group Policy: Core Protocol specified in [MS-GPOL] allows Group Policy clients to discover and retrieve the policy settings of Group Policy extensions that are configured by Group Policy administrators. These settings are persisted within GPOs that are assigned to **policy target** accounts in **Active Directory**, which can include computer accounts and user accounts. Each **Group Policy client** uses **Lightweight Directory Access Protocol (LDAP)**, via the **core Group Policy engine**, to access the GPOs in Active Directory and to determine which GPOs apply to it by consulting the **scope of management (SOM)** configuration. SOM is the collection of GPOs that apply to a set of policy targets, such as the computer and user accounts contained in sites, **domains**, or OUs that are associated with one or more GPOs.

On each Group Policy client, applicable GPOs are interpreted and acted upon by a **client-side extension (CSE)**. The CSE responsible for a given GPO is specified in a GPO Extension list. Extension lists are maintained by the **gPCMachExtensionNames** and **gPCUserExtensionNames** attributes of a GPO, the former of which contains **Group Policy extension GUID** pairs that apply to computer policy settings, and the latter of which contain Group Policy extension GUID pairs that apply to user policy settings. For Group Policy extensions that implement both a client and administrative side, these attributes contain a list of **GUID** pairs. The first GUID of each pair is referred to as the **client-side extension GUID (CSE GUID)**, while the second GUID of each pair is referred to as the **Administrative tool extension GUID**. The CSE GUID is typically used by a Group Policy client, via

the core Group Policy engine, to invoke a CSE (such as the Group Policy: Central Access Policies Extension defined in this specification) to facilitate the configuration of policy settings for that extension. The Administrative tool extension GUID is used by the **Administrative tool** to invoke the administrative side of an extension protocol during the policy administration process.

Whenever GPOs are created or updated, Group Policy fires the **Process Group Policy** event, as specified in section [3.2.4.1](#), which notifies Group Policy clients of a change in **Group Policy** by delivering a list of applicable GPOs. For each GPO, the Group Policy client consults the Extension lists of the GPO to discover the CSE GUIDs that indicates which CSE on the Group Policy client will handle the GPO. The Group Policy client then invokes the CSE to handle the policy configuration that is specified by the GPO.

A CSE uses GPO metadata to locate and retrieve settings that are specific to the CSE, and does so in a manner that is specific to that CSE. After the CSE-specific settings are retrieved, the CSE uses those settings to configure policy settings on Group Policy client computers.

For additional background information about Group Policy, refer to the Group Policy Protocols Overview document [\[MS-GPOD\]](#).

1.3.2 Central Access Policies Protocol Extension Overview

CAP settings identify authorization policies that are defined in **Active Directory**. More specifically, CAP settings contain the identifiers of authorization policies that are to be configured on **Group Policy client** computers for centralized control of user access to resources. An authorization policy is specified by a **central access rule (CAR)** that exists within a **CAP object**. The Group Policy: Central Access Policies Extension enables these authorization policies, specified within CAP settings, to be applied by authorization routines [\[MS-DTYP\]](#) section 2.5.3.2 on Group Policy client computers.

The general sequence in which CAPs are implemented is as follows:

- Author CAPs in Active Directory with an appropriate tool. CAP objects contain one or more central access rules (CARs), which in turn specify an authorization policy that defines how access to resources is controlled.
- Target specific Group Policy client computers for CAP application through GPO configuration and assignment.
- Invoke the CSE to populate the client-side ADM with CAP configuration data.
- Apply CAPs to individual Group Policy client resources (by a local resource administrator).
- Enforce CAP authorization rules on Group Policy client computers.

When a user attempts to access resources that have a CAP that was applied via access to client-side ADM values, the CAP authorization rules are enforced.

1.3.2.1 Central Access Policy Administration

Policy administration is driven by an **Active Directory** administrator and a **Group Policy administrator**. The administration of central access policies involves creating a **CAP object** and associating it with one or more GPOs.

Creating CAPs — An Active Directory administrator authors CAPs in Active Directory by using an administrative interface that can define authorization policies, such as an Active Directory Administrative Console. The **schema** for a CAP object is specified in [\[MS-ADSC\]](#) section 2.97 and the schema for the object's attributes is specified in [\[MS-ADA2\]](#) sections 2.123 through 2.129.

Configuring GPOs — Group Policy administrators configure CAP settings in Group Policy by:

- Using an **Administrative tool** to create or edit GPOs in Active Directory.
- Associating computer accounts with one or more GPOs.
- Specifying the CAPs for the computer accounts with which one or more GPOs is associated.

The administrative side of the Group Policy: Central Access Policies Extension interacts with the CAP policy file through an implementation-specific Administrative tool, such as the Group Policy Management Console. When the administrative-side extension is invoked by the Administrative tool, the Group Policy administrator can either create a new policy or retrieve and edit an existing one. If the Group Policy administrator is working with a new CAP policy, then he or she will create and configure a new GPO in Active Directory, which includes associating the GPO with one or more CAP objects and setting the GPO's **GPCFileSysPath** attribute to specify the **Group Policy file share** location where CAP policy settings are to be stored. If the Group Policy administrator is retrieving an existing policy, the GPO data is read and displayed by the Administrative tool and policy settings can then be modified as required. After the Group Policy administrator creates or modifies policy settings, the changes are propagated back into the logical component of the GPO and to the policy file on the Group Policy file share, via LDAP and a file access protocol, respectively. [<1>](#)

1.3.2.2 Central Access Policy Configuration Process

Group Policy clients are notified of changes in **Group Policy** when Group Policy fires the **Process Group Policy** event (section [3.2.4.1](#)).

The CSE of the Group Policy: Central Access Policies Extension protocol does not directly apply CAPs to **Group Policy client** computers; rather, it provides the configuration process that populates the client-side ADM. In turn, the ADM provides accessibility to the state required for the initial application and update of CAPs on Group Policy client computers via client-side administrative tools. These tools are run by a local resource administrator when he/she is ready to apply or update CAPs on Group Policy client computers.

Note In Group Policy, the periodic application of policy is triggered by the **core Group Policy engine** at regular refresh intervals, which is known as background **policy application**. This is different from the manual application of CAPs that is initiated by a local resource administrator.

To facilitate the CAP configuration process, CAP settings are retrieved by the CSE of the Group Policy: Central Access Policies Extension protocol following the trigger of the **Process Group Policy** event. The CSE uses LDAP to access the GPOs in **Active Directory** that contain the identifier-attributes that specify the location of CAP data, along with the file access protocol location where the policy settings are stored. The CAP configuration process on Group Policy client computers is then completed when the CSE performs the following:

- Retrieves the policy file containing the policy settings from the **Group Policy file share** via file access protocol sequences.
- Parses the file contents to obtain the LDAP distinguished names (DNs) of applicable **CAP objects**.
- Invokes LDAP to retrieve the authorization rules contained in the CAP objects in Active Directory.
- Populates the client-side ADM to maintain the state that enables the subsequent manual application of CAPs on Group Policy client computers.

Authorization policies are manually applied on a Group Policy client computer, such as a file server, by a local resource administrator with the use of an administrative tool. Following the application of CAPs, a Group Policy client is authorized to provide access to specific resources that are identified by the CAPs. For details on how CAPs are evaluated during the authorization process, refer to [\[MS-DTYP\]](#) section [MS-DTYP] section 2.5.3.2.

1.4 Relationship to Other Protocols

The Group Policy: Central Access Policies Extension depends on the Group Policy: Core Protocol specified in [\[MS-GPOL\]](#), to provide a list, via LDAP, of GPOs that apply to **policy target** accounts. This protocol also depends on LDAP for retrieving CAPs. The Group Policy: Central Access Policies Extension also transmits **Group Policy** settings and instructions between the **Group Policy client** and the **Group Policy file share** by reading and writing files via a file access protocol.

Note For an overview of file access concepts, see [\[MS-FASOD\]](#).

The following diagram illustrates the protocol relationships.

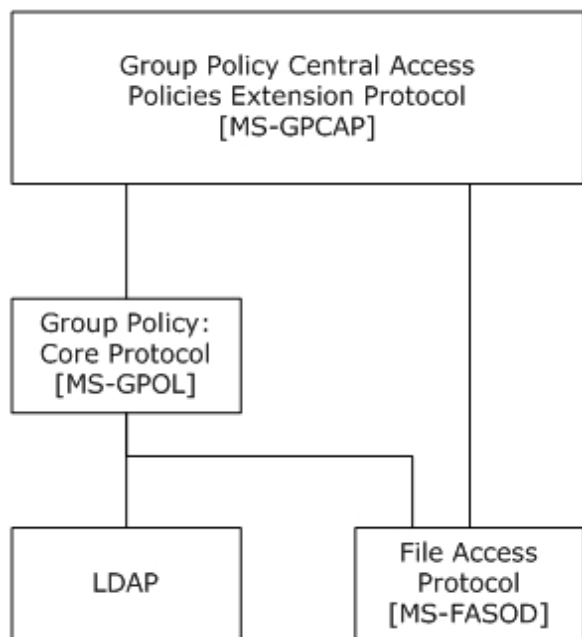


Figure 1: Group Policy: Central Access Policies Extension protocol relationships

1.5 Prerequisites/Preconditions

The prerequisites for the Group Policy: Central Access Policies Extension are identical to those specified in [\[MS-GPOL\]](#) section 1.5, in addition to the following:

- A valid **CAP object** exists in **Active Directory**.
Note The schema requirements for CAP objects are specified in section [2.3](#).
- The **Group Policy server** is a read/write **domain controller (DC)**.
- The **Group Policy client** is capable of discovering and communicating with the Group Policy server and can connect with Active Directory, as defined in [MS-GPOL] section 3.2.5.1.1.
- The **Administrative tool** is capable of discovering and communicating with the Group Policy server and can connect with Active Directory, as defined in [MS-GPOL] section 3.2.5.1.1.

1.6 Applicability Statement

The Group Policy: Central Access Policies Extension is only applicable within Group Policy.

1.7 Versioning and Capability Negotiation

None.

1.8 Vendor-Extensible Fields

None.

1.9 Standards Assignments

Standard assignments for the Group Policy: Central Access Policies Extension consist of a CSE GUID and an **Administrative tool extension GUID**, as specified in [\[MS-GPOL\]](#) section 1.8. The following table contains the assignments.

Parameter	Value
CSE GUID	{16be69fa-4209-4250-88cb-716cf41954e0}
Administrative tool extension GUID	{22b007da-4935-4079-9ec5-9c81507cc714}

2 Messages

2.1 Transport

The Group Policy: Central Access Policies Extension requires remote access to policy files, as specified in [\[MS-GPOL\]](#) section 2. All messages specified in section [2.2](#) of this specification MUST be exchanged via a file access protocol between the **Group Policy client** and **Group Policy server** and between the **Administrative tool** and the Group Policy server, assuming that the **Group Policy file share** is located on the Group Policy server (section [1.3](#)). [<2>](#)

The **core Group Policy engine** MUST use this protocol's CSE GUID and **Administrative tool extension GUID** values to invoke the client or administrative side of this protocol, respectively, which in turn invoke **LDAP** to access GPOs that require processing by this protocol.

2.2 Message Syntax

Messages exchanged in Group Policy: Central Access Policies Extension processes carry CAP policy file data that is transferred via file access sequences. This protocol is driven through the exchange of these messages, as specified in section [3](#).

2.2.1 Namespaces

None.

2.2.2 Central Access Policy File Message Format

All CAP policy files processed by the Group Policy: Central Access Policies Extension are **UTF-8** encoded and are based on the following file syntax.

```
InfFile = UnicodePreamble VersionPreamble Sections
UnicodePreamble = *("[Unicode]" LineBreak "Unicode=yes"
    LineBreak)
VersionPreamble = "[Version]" LineBreak "Signature="
    DQUOTE "$Windows NT$" DQUOTE LineBreak "Revision=1" LineBreak
Sections = Section / Section Sections
Section = Header Settings
Header = "[" HeaderValue "]" LineBreak
HeaderValue = stringWithSpaces
Settings = Setting / Setting Settings
Setting = DQUOTE Value DQUOTE LineBreak
Value = String
```

The preceding syntax is in the **Augmented Backus-Naur Form (ABNF)** grammar, as specified in [\[RFC4234\]](#), and is augmented by the following rules.

```
LineBreak = CRLF
StringWithSpaces = String / String Wsp stringWithSpaces
QuotedString = DQUOTE *(%x20-21 / %x23-7E) DQUOTE
Wsp = *WSP
ALPHANUM = ALPHA / DIGIT
```

Each Value string MUST be a valid LDAP distinguished name, as defined in [\[MS-ADTS\]](#) section 3.1.1.3.1.2.

Note CAP policy files are stored as .inf files in a subfolder (section [3.1.5.1](#)) of the Machine subdirectory in the **Group Policy Object (GPO) path**.

2.2.3 Central Access Policy ID Setting

This section defines settings that Group Policy administrators use to configure CAP identifiers. These settings identify the central access policies that are used by the **Group Policy client** extension for the configuration of access control policies for Group Policy client computer resources.

HeaderValue	Purpose
CAPS	This value MUST contain one or more settings that describe the LDAP DN's of CAP objects .

```
Header = "[" HeaderValue "]" LineBreak
HeaderValue = "CAPS"
Settings = Setting / Setting Settings
Setting = DQUOTE Value DQUOTE LineBreak
Value = String
```

Each Value string MUST be a valid LDAP distinguished name, as defined in [\[MS-ADTS\]](#) section 3.1.1.3.1.2. This distinguished name identifies a CAP object, as defined in [\[MS-ADSC\]](#) section 2.97. The Group Policy: Central Access Policies Extension uses the distinguished name to look up the CAP object in **Active Directory** and configure its settings in the Group Policy client computer ADM.

2.3 Directory Service Schema Elements

The Group Policy: Central Access Policies Extension accesses the directory service **schema** class and attributes listed in the following table.

For the syntactic specifications of the following schema classes and attributes, refer to **Active Directory Domain Services (AD DS)** ([\[MS-ADSC\]](#) sections 2.96 through 2.99; and [\[MS-ADA2\]](#) sections 2.123 through 2.129).

Class	Attributes
msAuthz-CentralAccessPolicy	msAuthz-CentralAccessPolicyID
	msAuthz-MemberRulesInCentralAccessPolicy
	msAuthz-MemberRulesInCentralAccessPolicyBL
msAuthz-CentralAccessRule	msAuthz-EffectiveSecurityPolicy
	msAuthz-LastEffectiveSecurityPolicy
	msAuthz-ProposedSecurityPolicy
	msAuthz-ResourceCondition

3 Protocol Details

3.1 Central Access Policies Protocol Administrative-Side Extension Details

The administrative side of the Group Policy: Central Access Policies Extension participates in authoring CAP settings via GPO configuration, as specified in section [1.3.2.1](#). A **central access policy (CAP)** MUST be stored as a text file in ".inf" file format, as specified in section [2.2](#). The CAP file MUST be stored in a location that is accessible via RFA sequences.

3.1.1 Abstract Data Model

None.

3.1.2 Timers

None.

3.1.3 Initialization

When the administrative-side extension of this protocol is invoked, it MUST obtain a **computer-scoped Group Policy Object path** from the **gPCFileSysPath** attribute of a GPO via the **core Group Policy engine**, as specified in [\[MS-GPOL\]](#) section 2.2.4. It MUST then perform the processing instructions specified in section [3.1.5.1](#).

Note The administrative-side extension of this protocol does not maintain any local state and therefore does not require local state variables nor any subsequent variable initialization. The administrative-side extension loads all the settings specified in section [2.2](#) into memory.

3.1.4 Higher-Layer Triggered Events

The following higher-layer triggered events occur in response to the indicated trigger conditions:

Event	Trigger condition
Load Policy	The Administrative tool is initialized or the Group Policy administrator loads a CAP.inf file (section 3.1.5.1) for the Group Policy: Central Access Policies Extension.
Update Policy	The Group Policy administrator updates any policy setting value (section 3.1.5.2) for the Group Policy: Central Access Policies Extension.
Delete Setting Value	The Group Policy administrator deletes any policy setting value (section 3.1.5.3) for the Group Policy: Central Access Policies Extension.

3.1.5 Message Processing Events and Sequencing Rules

The administrative-side extension of this protocol invokes a file access protocol to read extension-specific data from the CAP.inf policy file that is stored in the **Group Policy file share** and then passes that information to the **Administrative tool**. The tool provides an interface that displays the current extension settings to the **Group Policy administrator**. If the Group Policy administrator modifies the existing extension settings, the administrative-side extension invokes the **Update Policy** event (section [3.1.5.2](#)).

If a CAP.inf policy file does not yet exist, the administrative-side extension MUST create it in the remote location specified in section [3.1.5.1](#) when the Group Policy administrator saves the initial CAP policy configuration data.

Note File names and paths are regarded as case-insensitive. If the File Open or File Write operations fail, the **Administrative tool extension** MUST provide an error indication to the Group Policy administrator that the operation failed.

Note Each time a policy file is created, modified, or deleted, the administrative-side extension MUST invoke the **Group Policy Extension Update** event, as specified in [\[MS-GPOL\]](#) section 3.3.4.4.

3.1.5.1 Load Policy

The **Load Policy** event occurs when the **Group Policy administrator** invokes the administrative-side extension of this protocol and its dynamic linked library (DLL) is loaded into the **Administrative tool**. After the administrative-side extension is loaded, it MUST obtain the **computer-scoped Group Policy Object path** from the **gPCFileSysPath** attribute of a GPO via the **core Group Policy engine**, as specified in [\[MS-GPOL\]](#) section 2.2.4. The extension MUST then attempt to retrieve data from any existing CAP.inf file stored in the following location:

<gpo path>\Machine\Microsoft\Windows NT\CAP\

where <gpo path> is the Universal Naming Convention (UNC) path to the physical file share location where the CAP policy file is stored. For example: "\\<dns domain name>\<GP FS-name>\<dns domain name>\policies\<gpo guid>", where <dns domain name> is the **DNS domain name**, and <gpo guid> is a GPO **GUID**.

Note The core Group Policy engine invokes a file access protocol on behalf of the administrative-side extension when retrieval of CAP data is required.

At this point, the file access protocol MUST perform the file read and parse operations specified in section [3.2.5.2](#). If the attempt to read the CAP.inf file fails, an error MUST be logged and processing MUST be stopped.

3.1.5.2 Update Policy

The **Update Policy** event occurs when the **Group Policy administrator** updates the policy settings in the file system component of a GPO by using the **Administrative tool**. When policy settings are modified, the state of the GPO MUST be updated via the following **Update Policy** message sequence:

1. File access *File Open* sequence:

The **Administrative tool extension** MUST first invoke the **core Group Policy engine** to obtain the <gpo path>, as specified in [\[MS-GPOL\]](#) section 2.2.4, to locate the CAP.inf file.

The file access *File Open* operation MUST request write permissions and MUST create the file if it does not exist. If it does not exist, the operation MUST attempt to write a CAP.inf file to the following location:

<gpo path>\Machine\Microsoft\Windows NT\CAP\

If the *File Open* request returns an implementation-specific failure status, the entire Group Policy: Central Access Policies Extension sequence MUST be terminated.

2. File access *File Write* sequences:

The Administrative tool extension MUST perform a series of file writes to overwrite the contents of the opened file with new policy settings. These file writes MUST continue until the entire file is written or an error is encountered.

If the *File Write* request returns an implementation-specific failure status, the entire Group Policy: Central Access Policies Extension sequence MUST be terminated.

3. File access *File Close*:

The Administrative tool extension **MUST** then issue a *File Close* operation.

4. Providing that no failures occurred, the Administrative tool extension **MUST** invoke the **Group Policy Extension Update** event ([MS-GPOL] section 3.3.4.4).

3.1.5.3 Delete Setting Value

The **Delete Setting Value** event occurs when the **Group Policy administrator** deletes a **policy setting** value. When a policy setting value is deleted, the setting is removed from memory and the processing described in section [3.1.5.2](#) **MUST** be performed.

3.1.6 Timer Events

None.

3.1.7 Other Local Events

None.

3.2 Central Access Policy Configuration Client-Side Extension Details

The CSE of the Group Policy: Central Access Policies Extension interacts with Group Policy as specified in [\[MS-GPOL\]](#) section 3.2. The CSE **MUST** retrieve the **central access policy (CAP)** (section [3.2.5](#)) and modify the appropriate part of the ADM for each element in the policy, as specified in this section.

3.2.1 Abstract Data Model

This section defines a conceptual model of possible data organization that an implementation maintains to participate in this protocol. The described organization is provided to explain how the protocol behaves. This document does not mandate that implementations adhere to this model as long as their external behavior is consistent with what is described in this document.

3.2.1.1 Policy Setting State

The persistent state configured by the CSE of this protocol is specified herein. The location for storing this state is implementation-specific.

Note The abstract interface notation (Public) for an ADM element indicates that the data element can be directly accessed from outside this protocol.

CentralAccessPolicyDNList: A persistent list of string-valued data elements. The string value of each element is the **LDAP** distinguished name of an existing **CAP object**.

CentralAccessPoliciesList (Public): A persistent list of **CentralAccessPolicy** objects.

CentralAccessPolicy: A structure data type that contains the following fields.

Field name	Description
CAPID	A security identifier (SID) , as specified in [MS-DTYP] section 2.4.2, that identifies the CentralAccessPolicy object.
CentralAccessPolicyDN	The LDAP distinguished name of the CentralAccessPolicy object.
CentralAccessPolicyRulesList	A list of CentralAccessPolicyRule objects.

CentralAccessPolicyRule: A structure data type that contains the following fields.

Field name	Description
EffectiveCentralAccessPolicy	A data element of type CentralAccessPolicyCondition containing the effective access policy for the CentralAccessPolicyRule . The schema class for a CentralAccessPolicyRule is defined in [MS-ADSC] section 2.98.
StagedCentralAccessPolicy	A data element of type CentralAccessPolicyCondition containing the staged access policy for the CentralAccessPolicyRule . The schema class for a CentralAccessPolicyRule is defined in [MS-ADSC] section 2.98.

CentralAccessPolicyCondition: A structure data type that contains the following fields.

Field name	Description
AppliesToPredicate	An ACCESS_ALLOWED_CALLBACK_ACE value ([MS-DTYP] section 2.4.4.6) that contains the condition that defines the scope of the resources to which the CentralAccessPolicyEntry data element applies.
AccessCondition	A security descriptor value ([MS-DTYP] section 2.4.6) that contains the access condition for the CentralAccessPolicyEntry data element.

3.2.2 Timers

None.

3.2.3 Initialization

When the CSE of the Group Policy: Central Access Policies Extension protocol is invoked by Group Policy, and a list of one or more applicable GPOs is provided for updates, the CSE MUST do the following:

1. Locate all the CAP.inf policy files specified by the metadata of each GPO.
Central access policy files MUST be located by performing the tasks outlined in [\[MS-GPOL\]](#) section 3.2.5.1, which includes sending the appropriate LDAP **SearchRequests** ([\[RFC2251\]](#) section 4.5) to query GPOs in **Active Directory**.
2. Copy the policy files to the **Group Policy client** computer.
3. Read the policy files to determine the location of **CAP objects** in Active Directory.
4. Configure the client-side ADM, as specified in section [3.2.5](#).

Note The policy files MUST be copied and read by using file access sequences.

3.2.4 Higher Layer Triggered Events

The CSE of this protocol receives the following higher-layer triggered event:

- **Process Group Policy** (section [3.2.4.1](#)).

3.2.4.1 Process Group Policy

The CSE of Group Policy: Central Access Policies Extension protocol implements the **Process Group Policy** abstract event interface, as specified in [MS-GPOL] section 3.2.4.1. The CSE does not make use of the *Deleted GPO list*, *SessionFlags*, or the *SecurityToken* logical parameters of the event; rather, it only requires the *New or Changed GPO list* parameter. When the event is triggered, the CSE MUST perform the actions specified in section 3.2.5.

3.2.5 Message Processing Events and Sequencing Rules

3.2.5.1 Client-Side Extension Invocation

The CSE of the Group Policy: Central Access Policies Extension protocol MUST be invoked by the **Process Group Policy** event specified in [MS-GPOL] section 3.2.5.1.10 whenever applicable GPOs require processing on the **Group Policy client**, as determined by the **policy application** process specified in [MS-GPOL] section 3.2.5.1. When this occurs, the CSE of this protocol MUST perform the actions specified in sections 3.2.5.2 and 3.2.5.3.

3.2.5.2 Client-Side Extension Sequences

When invoked by the **Process Group Policy** event, the CSE attempts to retrieve the list of applicable GPOs from the *New or changed GPOs* logical parameter of the event. The CSE MUST then iterate through this list and locate and retrieve the central access policy file (CAP.inf) in the path specified by the **gPCFileSysPath** attribute of each GPO. For each GPO, one file with the format specified in section 2.2 MUST be copied from the **Group Policy file share** to the local computer.

For each GPO, the CSE of this protocol MUST generate the following file access sequences when processing each CAP.inf file:

Sequence	Description
File Open	The CSE MUST attempt to open the file specified in the following location: <scoped gpo path>\Microsoft\Windows NT\CAP\cap.inf.
File Read	Until an error occurs, one or more file reads MUST be performed to read the entire contents of the opened file.
File Close	A file close operation MUST be performed.

Note If any file cannot be read, the CSE MUST log information about the failure and continue to process CAP.inf files specified by other GPOs.

Each file MUST be parsed according to the format specified in section 2.2. If the file does not conform to the specified format, the entire operation for that file MUST be ignored. If the file does conform to the specified format, each distinguished name Value specified in Settings in the CAP.inf file (section 2.2.2) MUST be added to the **CentralAccessPolicyDNList** ADM element described in section 3.2.1.1.

3.2.5.3 Policy State Configuration

After all the distinguished name values are retrieved, the CSE MUST perform the following steps for each entry in the **CentralAccessPolicyDNList** ADM element. If any LDAP operations fail, the corresponding distinguished name entry MUST be ignored.

1. Perform an LDAP bind to the **CAP object** in **Active Directory** by using the LDAP distinguished name specified by the **CentralAccessPolicyDNList** ADM element entry value, as created in section 3.2.5.2.

2. Create a new **CentralAccessPolicy** ADM element and add it to the **CentralAccessPoliciesList** ADM element. Populate the fields of this element as follows:
 - Set the value of the **CAPID** field of this new **CentralAccessPoliciesList** ADM element entry to the value obtained by performing an LDAP read of the **msAuthz-CentralAccessPolicyID** attribute on the object that was bound to in step 1.
 - Set the **CentralAccessPolicyDN** ADM field value of this new entry to the LDAP distinguished name of the CAP object that was bound to in step 1.
 - Create a new **CentralAccessPolicyRulesList** ADM structure.
 - Perform an LDAP read of the **msAuthz-MemberRulesInCentralAccessPolicy** attribute of the CAP object bound to in step 1 to obtain the list of DN's of CAR object rule entries. If this list is empty, ignore this entry.
 - For each CAR object distinguished name in the list obtained in step 2 bullet 4, create a new **CentralAccessPolicyRule** ADM structure, perform an LDAP bind on the CAR object by using the distinguished name, and then do the following:
 - Set the value of the **AppliesToPredicate** data element field of the **EffectiveCentralAccessPolicy** data element field of the **CentralAccessPolicyRule** structure to the binary equivalent of the security descriptor definition language (SDDL) ([\[MS-DTYP\]](#) section 2.5.1) string value obtained by performing an LDAP read of the **msAuthz-ResourceCondition** attribute of the CAR object bound to in step 2 bullet 5.
 - Set the value of the **AccessCondition** ADM element field of the **EffectiveCentralAccessPolicy** ADM element field of the **CentralAccessPolicyRule** structure to the binary equivalent of the SDDL string value obtained by performing an LDAP read of the **msAuthz-EffectiveSecurityPolicy** attribute of the CAR object bound to in step 2 bullet 5.
 - Set the value of the **AppliesToPredicate** data element field of the **StagedCentralAccessPolicy** data element field of the **CentralAccessPolicyRule** structure to the binary equivalent of the SDDL string value obtained by performing an LDAP read of the **msAuthz-ResourceCondition** attribute of the CAR object bound to in step 2 bullet 5.
 - Set the value of the **AccessCondition** data element field of the **StagedCentralAccessPolicy** data element field of the **CentralAccessPolicyRule** structure to the binary equivalent of the SDDL string value obtained by performing an LDAP read of the **msAuthz-ProposedSecurityPolicy** attribute of the CAR object bound to in step 2 bullet 5.
 - Add the populated **CentralAccessPolicyRule** ADM structure created in step 2 bullet 5 to the **CentralAccessPolicyRulesList** ADM structure created in step 2 bullet 3.
3. Add the **CentralAccessPolicyRulesList** ADM structure created in step 2 bullet 3 to the **CentralAccessPolicy** ADM structure created in step 2.

3.2.6 Timer Events

None.

3.2.7 Other Local Events

None.

4 Protocol Examples

4.1 Example of a CAP.inf File

```
[Version]

Signature="$Windows NT$"

[CAPS]

"CN=LCA Document Access,CN=Central Access Policies,CN=Claims
Configuration,CN=Services,CN=Configuration,DC=DMM-
CBACDOM,DC=nttest,DC=microsoft,DC=com"

"CN=MSIT Corporate Standard Access Policy,CN=Central Access Policies,CN=Claims
Configuration,CN=Services,CN=Configuration,DC=DMM-
CBACDOM,DC=nttest,DC=microsoft,DC=com"
```

5 Security

5.1 Security Considerations for Implementers

A **central access policy** defines an authorization policy that controls access to resources. Write permissions on central access policies enable a user to modify the authorization policy. Central access control policies are designed to be managed centrally and not be edited on client computers. Therefore, it is important to store central access control policies on client computers in secure locations to which only system processes have access.

5.2 Index of Security Parameters

None.

6 Appendix A: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include updates to those products.

- Windows Server 2012 operating system
- Windows Server 2012 R2 operating system
- Windows Server 2016 operating system
- Windows Server operating system
- Windows Server 2019 operating system
- Windows Server 2022 operating system
- Windows Server 2025 operating system

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

<1> [Section 1.3.2.1](#): In Windows, Group Policy and this protocol extension use file access services protocols (see [\[MS-FASOD\]](#) for file access operations).

<2> [Section 2.1](#): In Windows, the **Group Policy file share** repository is a **system volume (SYSVOL)** share on the **Group Policy server**.

7 Change Tracking

This section identifies changes that were made to this document since the last release. Changes are classified as Major, Minor, or None.

The revision class **Major** means that the technical content in the document was significantly revised. Major changes affect protocol interoperability or implementation. Examples of major changes are:

- A document revision that incorporates changes to interoperability requirements.
- A document revision that captures changes to protocol functionality.

The revision class **Minor** means that the meaning of the technical content was clarified. Minor changes do not affect protocol interoperability or implementation. Examples of minor changes are updates to clarify ambiguity at the sentence, paragraph, or table level.

The revision class **None** means that no new technical changes were introduced. Minor editorial and formatting changes may have been made, but the relevant technical content is identical to the last released version.

The changes made to this document are listed in the following table. For more information, please contact dochelp@microsoft.com.

Section	Description	Revision class
6 Appendix A: Product Behavior	Added Windows Server 2025 to the list of applicable products.	Major

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